

CO-2 CASE STUDY

STUDENT CASE STUDY EXECUTION REPORT

Department: FED

Subject: DSA-2

Academic Year: I-III Semester (2025-2026)

Faculty Name: Ms. G.Mamatha

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Branch: CSE

Course Outcome:

CO-2 B-Tree

Case Study Topic:

Efficient Student Database Management using B-Tree

Work Done Summary:

In this case study, student records are managed using a B-Tree data structure. B-Tree is used to organize and store large volumes of student information efficiently. Student IDs act as indexing keys, enabling quick insertion, searching, and retrieval of records. The B-Tree structure maintains balance automatically and reduces access time, making it suitable for handling large educational databases.

Implementation Details:

1. Student records are stored using a B-Tree structure.
2. Student ID is used as the primary indexing key.
3. New student records are inserted into the appropriate B-Tree node.
4. When a node becomes full, node splitting is performed automatically.
5. Records remain sorted within each node.
6. Search operations quickly locate student information.
7. B-Tree maintains balanced structure during insertions.
8. Large student databases are managed efficiently.
9. The system supports fast retrieval and organized storage of records.
10. The balanced structure improves database performance.

Result : Screenshots / Output

B-Tree Structure:

```
[103, 106]
 /      |      \
[100,101,102] [104,105] [107,108,110]
```

Student Records:

```
101 (Anjali)
102 (Kiran)
103 (Charitha)
105 (Rahul)
110 (Priya)
```

Searching Student ID: 105

Record Found Successfully
Student Name: Rahul

B-Tree Operations Completed Successfully

```
Search Time : O(log n)
Insertion Time : O(log n)
Deletion Time : O(log n)
```

1. Student records were inserted successfully into the B-Tree.
2. Search operations retrieved student information efficiently.
3. Node splitting maintained the balanced structure.
4. Large datasets were managed effectively.
5. Search Operation: **$O(\log n)$**
6. Insertion Operation: **$O(\log n)$**
7. Deletion Operation: **$O(\log n)$**
8. B-Tree improved database indexing and retrieval performance.

GitHub Link: <https://github.com/devicharithab-del/DSA-2-CO2>

Student Signature

Faculty Signature